CYBR-2200 Network Analysis

WEEK 6 — Network Monitoring, Traffic Statistics, and Detection

📊 Gradebook Configuration

- **Assignment Name:** Week 6 Lab: Network Monitoring & Traffic Statistics
- **Category:** Weekly Labs
- **Point Value: 90 points**
- **Counts Toward:** Weekly attendance & engagement
- **Submission Type:** File upload (DOCX or PDF)
- **GitHub:** ☑ **Recommended** (not required)

🎯 Assignment Purpose

This lab teaches you how to:

- Analyze network traffic at scale
- Use statistics to identify patterns
- Detect high-risk or unusual behavior
- Think like a security operations analyst

This mirrors how analysts work in real environments.

📝 What Students Must Do (Explicit)

Students must analyze **all PCAP files provided for Week 6** and complete **every section** of the worksheet:

Part A — Open PCAP Files

- Successfully open each PCAP in Wireshark

Part B — Traffic Overview

- Use:
- Protocol Hierarchy
- Conversations
- Endpoints
- Identify dominant protocols, hosts, and ports

	Protocol Hierarchy	Conversations	Endpoints	Identify dominant protocols, hosts, and ports.
2022-MTA-workshop-Quakbot-1-of-2.pcap	TCP (96.6%) TLS (7.2%) Data (3.9%)	10.7.15.101>23.201.27.241 10.7.15.101>24.178.196.158 10.7.15.101>23.111.114.52	10.7.15.101 23.201.27.241 24.178.196.158	TCP (Port 443) /TLS1.2 The single internal system (10.7.15.101) is making repeated encrypted connections to a specific external IP (23.201.27.241) in a pattern consistent with command-and-control traffic malware.
2022-MTA-workshop-Quakbot-2-of-2.pcap	TCP = 89.2% of packets TLS = 23.7% of packets TLS = 86.1% of bytes	10.5.17.151>23.196.167.186 10.5.17.151>193.29.13.216 10.5.17.151>23.111.114.52 10.5.17.151>83.106.215.197	10.5.17.151 23.196.167.186 193.29.13.216	HTTPS encrypted through TCP/TLS on Port 443 again is consistent with a Quakbot malware C2 infection.

Part C — Pattern Identification

- Identify repeated connections or trends
- Explain why certain traffic patterns are significant
- Identify potential indicators of malicious activity

🔑 **Quakbot 1-of-2**

This PCAP analysis shows that the IP 10.7.15.101 (internal host) repeatedly communicated with the IP 23.201.27.241 (external host) over TCP port 443 using TLS1.2. The Protocol Hierarchy showed a majority proportion of the total bytes sent were encrypted such as HTTPS. The encryption sessions last only seconds to various external hosts over encrypted sessions. This is consistent with the Quakbot beaconing C2 infrastructure to external IPs.

Quakbot 2-of-2

The second capture shows the IP 10.5.17.151 (internal host) produced the majority of the outbound traffic to multiple sessions over TCP to various external IP addresses. The significance of this is their length, lasting between 600 and 3300 seconds. This is consistent with the encrypted TLS traffic shown with a Quakbot malware infection though the internal IPs Command-and-Control infrastructure.

Part D — Analyst Assessment

- Explain how statistics support detection better than packet-by-packet review

  By utilizing statistical analysis, one is able to more effectively use algorithms, filters, and patterns to detect anomalies on a much faster scale than moving packet by packet through a system. For example, Quakbot packet 2 had 8000+ packets and would have taken 16.67 days working 8 hours a day to see the first 8000 packets if we took one minute on investigating each packet. I was able to use the statistical analysis to shave that down to seconds. These grouped views allows more anomalies to be detected such as the Quakbot. You are able to clearly see the command-and-control communication, repeated login attempts through brute force attacks, repeated connections, and other anomalies which gives analysts and incident response personnel a chance at being able to contain and remediate the infectious computer at a much faster rate.

📤 Submission Requirements

Students submit:

- **One completed worksheet**
- DOCX or PDF format
- Uploaded to Blackboard

🚫 No PCAP uploads
🚫 Screenshots optional only

AI**

SAGE AI was used to help read and understand the PCAPs.